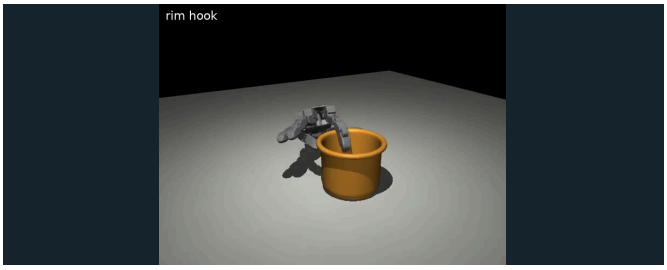
RESEARCH BRIEF, OCTOBER 2026

Learning to grasp with vision and touch under shape uncertainty

An object's hidden geometry may affect how it should be grasped. POgrasp asks whether contact can update a belief over plausible shapes and help a dexterous hand choose a better action.

> **Current status.** The grasping and evaluation pipeline is in place. Existing experiments use privileged object information. Integrating a visual prior with contact-based belief updates is proposed work.

Recorded behaviors



Open cup. A learned rim grasp on a training object.



Lidded cup. A learned handle-directed approach with the true handle direction supplied.

These are selected rollouts from the same two-object policy checkpoint. They show learned behaviors, not generalization or average performance. The accompanying project page contains the original videos, including an earlier body-pinch shortcut.

Shape-input diagnostics

Separate recorded comparisons	Provided	Zeroed
Can and open cup: shape-summary channels	32/32 each	32/32 each
Lidded cup: handle-direction channels	22/32	10/32

Each row compares the same checkpoint on the same 32 placements using mean actions. Success: a supported lift of 10 cm held for 2 seconds. The rows are different experiments; the images above are from a later checkpoint. Zeroing inputs can create unfamiliar observations, so the comparison shows input sensitivity rather than proof of general shape understanding.

An important negative result

An initial contact-history estimator did not reliably replace the true handle direction. It also received privileged object displacement and shape summaries. This result motivates testing informative contact and a visual prior; it does not establish that contact is generally insufficient.

Recorded September 2026. Prepared from dated research records; no new training or evaluation was run for this brief.

What the research will test

1. **Shape representation.** Preserve grasp-relevant detail while retaining multiple plausible shapes. Investigate equivariant encoders that keep intrinsic shape distinguishable from pose without losing reconstruction quality.
2. **Contact-based updates.** Connect a visual prior to evidence from contact. Test whether the policy gathers useful information and uses it in subsequent actions.
3. **Grasping benefit.** Compare vision, contact and their combination against matched baselines, including a student trained from a full-state teacher, with repeated seeds and new objects.

Why GPU support is needed

The proposed comparison requires repeated simulation-based training, baseline runs and evaluation. Shared lab GPU access is suitable for small pilots but makes sustained comparisons difficult to schedule. The initial object comparison is the minimum study; the broader evaluation depends on available resources.

What will be delivered

The intended outputs are a controlled comparison, experiment configurations, evaluation procedures and an analysis of where the approach helps or fails. Improved grasp performance is a hypothesis to test, not a promised result.

The work is intended for a research paper and a Master’s thesis, with CoRL as a target venue and ICRA another possibility depending on the final results. Code and permitted reproducibility materials are planned for release after paper publication.

Limits of the existing evidence

The pilots use a simulated DG-5F hand with a floating wrist and small trained object sets. They do not demonstrate a complete vision-and-touch policy, real-robot performance, an equivariant belief implementation or superiority over published baselines. Some successful lifts use a body-pinch shortcut, and the hand motions are not yet natural.

Experiment references

Shape-summary masking: internal record WALL_B_2026-09-17, B4 update 1620. Handle-direction masking: WALL_C_DESIGN_2026-09-19, B7 update 3660. Images: B10 update 4260. History-estimator findings: B12 v2, reviews 4620–4920. The accompanying evidence.json records the counts and media hashes. Original experiment artifacts are retained; this brief is not a fresh reproduction.